

Plotting with Matplotlib

A quick-reference guide for creating line, area, histogram, bar, pie, box, and scatter plots — both through Pandas' built-in plotting API and directly with Matplotlib.

PANDAS

MATPLOTLIB

PYTHON

Plotting Directly with Pandas

df.plot(kind=...)

Line Plot

Shows trends and changes over time



PANDAS FUNCTION

```
DataFrame.plot.line()  
DataFrame.plot(kind='line')
```

EXAMPLE

```
df.plot(x='year', y='sales', kind='line')
```

Area Plot

Displays data series as filled areas, showing the relationship between them



PANDAS FUNCTION

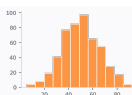
```
DataFrame.plot.area()  
DataFrame.plot(kind='area')
```

EXAMPLE

```
df.plot(kind='area')
```

Histogram

Displays bars representing the data count in each interval/bin



PANDAS FUNCTION

```
Series.plot.hist()  
Series.plot(kind='hist', bins=n)
```

EXAMPLE

```
s.plot(kind='hist', bins=10)  
df['age'].plot(kind='hist', bins=10)
```

Bar Chart

Displays data using rectangular bars



PANDAS FUNCTION

```
DataFrame.plot.bar()  
DataFrame.plot(kind='bar')
```

EXAMPLE

```
df.plot(kind='bar')
```

Pie Chart

Displays data as a circular plot divided into slices, representing proportions or percentages of a whole



PANDAS FUNCTION

```
Series.plot.pie()

Series.plot(kind='pie')

DataFrame.plot.pie(y, labels)

DataFrame.plot(kind='pie')
```

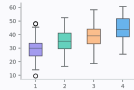
EXAMPLE

```
s.plot(kind='pie', autopct='%1.1f%%')

df.plot(x='Category', y='Percentage', kind='pie')
```

Box Plot

Displays the distribution of a dataset along with key statistical measures



PANDAS FUNCTION

```
DataFrame.plot.box()

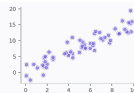
DataFrame.plot(kind='box')
```

EXAMPLE

```
df_can.plot(kind='box')
```

Scatter Plot

Uses Cartesian coordinates to display values for two variables



PANDAS FUNCTION

```
DataFrame.plot.scatter()

DataFrame.plot(x, y, kind='scatter')
```

EXAMPLE

```
df.plot(x='Height', y='Weight', kind='scatter')
```